

Fire Assignment 12

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1. ☒ BONUS Assignment

☐ How far could a new fire run in 8 hours?

Today, you'll simulate the probability that an ignition's 8-hour straight-line spread distance (km) exceeds a threshold of 20 km.

Replace each ____ placeholder (and any TODO comments) with working code or a short written answer. Run each section; be sure the requested objects appear in the Environment. Use the template ☐ Goal → ☐ Code → ☐ Checks → ☐ Explain.

When finished, save **BOTH** you R code script file and your .RData workspace and upload to Canvas. ____

1.1 Dataset

Wildfire managers need to know where wildfires are likely to occur and how intense they might be. The U.S. Forest Service (USFS) Wildfire Hazard Potential (WHP) is a raster geospatial product [Dillon, G., Gilbertson-Day, J., 2020](#) paired with Idaho-only subset data provided by Dr. Matt Williamson, an Associate Professor at Boise State University in Human-Environment Systems. Dr. Williamson specializes in spatial analysis using R and teaches the graduate level R course HES 505 each fall. To learn more about Dr. Williamson work, visit his [SPASES Lab Page](#).

Figure 1 shows the raster that was processed in R using packages terra, sf, tidyverse, tidyterra, tigris. The GeoTIFF is projected in USA_Contiguous_Albers Equal Area Conic (USGS version) (Albers; +proj=aea +lon_0=-96 +lat_1=29.5 +lat_2=45.5 +lat_0=23; meters) with a cell size of 240×240 m. Within Idaho, valid raster cells cover 216,315 km² ($\approx$ the state area); cells outside the state footprint within the raster's bounding box are NoData ($\approx$ 231,477 km²). WHP values in the subset range from 0 to 64,185.656 (mean 1,615.668, SD 4,119.674). Distributional summaries are: P25 = 130.484, median = 397.297, P75 = 1,097.703, P90 = 3,507.469, P95 = 7,753.109, P99 = 22,429.154 (all unitless). WHP's native grid is 270 m; the Idaho file's 240 m cell size indicates the subset was resampled prior to delivery. Higher index values indicate greater relative hazard; the layer is continuous. TIGER/Line state cartographic boundary for Idaho (retrieved via the R tigris package and reprojected to the raster CRS).

Formal dataset citation

Dillon, Gregory K.; Gilbertson-Day, Julie W. 2020. Wildfire Hazard Potential for the United States (270-m), version 2020. 3rd Edition. Forest Service Research Data Archive. <https://doi.org/10.2737/RDS-2015-0047-3>.

Idaho subset (whp_id.tif) prepared and provided by Matt Williamson, SPASES Lab, Boise State University.

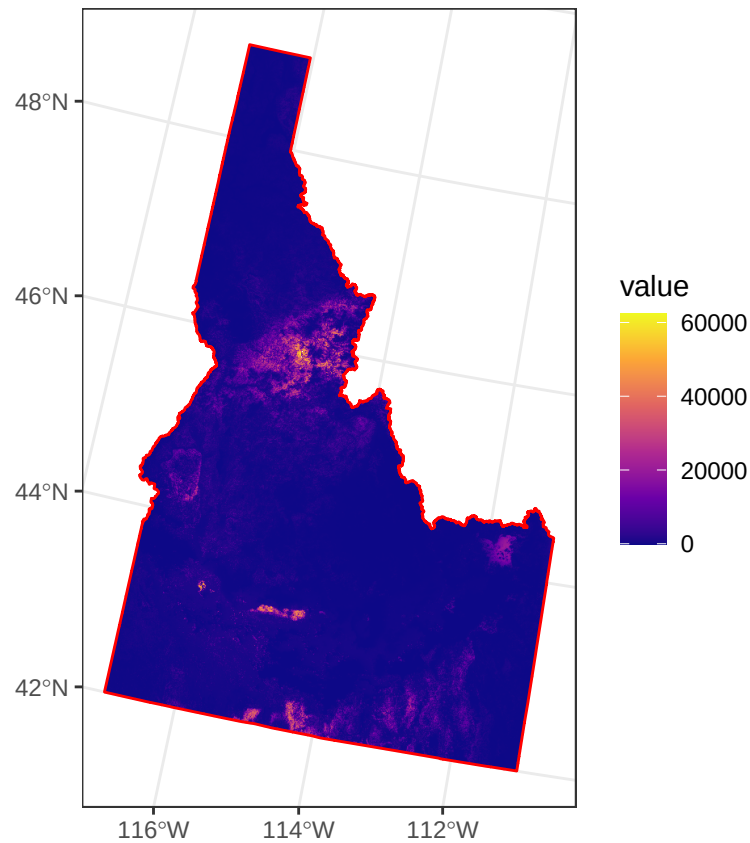


Figure 1.1: Figure 1. Idaho Wildfire Hazard Potential (WHP), continuous index (unitless) from the USFS WHP v2020 raster, clipped to the Idaho boundary. Colors use a viridis gradient (dark purple = lower WHP; yellow = higher WHP); values span approximately 0–64,186. Raster resolution is 240 m in an Albers Equal Area projection; NoData cells are transparent. Red outline shows the TIGER/Line state boundary reprojected to the raster CRS. Idaho subset prepared by Matt Williamson (SPASES Lab, Boise State University); original dataset: Dillon & Gilbertson-Day (2020), USFS Research Data Archive (DOI: 10.2737/RDS-2015-0047-3).

Table 1. summarizes the quantile-based wildfire hazard categories derived from Idaho’s continuous USFS WHP raster and the distance distributions used for an 8-hour run. Category lists seven bins (Very Low to Ultra) defined from the raster’s cumulative distribution; the WHP index itself is unitless. Probability is the statewide area share for each bin—expressed as a percent of valid raster area (NA excluded)—and is interpreted as the chance a random ignition falls in that category. Normal mean (km) and Normal SD (km) are the mean and standard deviation, in kilometers, for a zero-truncated normal distance model representing straight-line spread over 8 hours. Mean log and SD log are the parameters μ and σ of a lognormal distance model in which $\log(\text{distance in km}) \sim N(\mu, \sigma^2)$. Labels “Extreme” and “Ultra” denote upper-tail bins (approximately the 95–99% and 99–100% quantiles, respectively).

Parameters for this assignment were selected from the the following references. According to the Idaho Department of Environmental Quality (DEQ), the 2021 fire season saw 439 individual wildfires. A reasonable single-year ignition count for Idaho is ~ 450 (rounding the 2021 DEQ total of 439).

Category	Probability	Mean (km)	SD (km)	Mean log	SD log
Very Low	25	2.0	1.0	0.582	0.472
Low	25	3.0	1.0	1.046	0.325
Moderate	25	4.5	1.5	1.451	0.325
High	15	7.0	2.0	1.907	0.280
Very High	5	12.0	3.0	2.455	0.246
Extreme	4	18.0	4.0	2.866	0.220
Ultra (Top 1%)	1	28.0	6.0	3.310	0.212

1.2 Objective

Objective. Estimate short-term spread risk using simulation. Model each ignition’s 8-hour straight-line distance with a lognormal distribution. Do this two ways:

Statewide mixture: sample a WHP category using the Probability column in Table 1, then draw a distance from that category’s Mean log and SD log. An ignition critical if it distance ≥ 20 km

You will simulate 2,000 seasons, each with 450 ignitions. For each season, count how many ignitions are ≥ 10 km (major runs) and how many are critical under the rule. Summarize both expected counts and the probability the season exceeds a planning threshold (e.g., “ > 50 major runs” and “ > 10 critical runs”). Distance is used here as a proxy for severity; in practice agencies track area. Category probabilities approximate where ignitions occur and are a teaching simplification.

1.3 Lognormal Random Numbers

The lognormal model is used because most fires will move a modest distance, but a small fraction travel much farther. A lognormal model enforces non-negativite distance values and captures that right tail.

Use `rlnorm()` to generate lognormal random numbers.

```
?rlnorm # Check the help file for rlnorm()
```

□ The SYNTAX: `rlnorm(n, meanlog = 0, sdlog = 1)`

- `n` is the number of random numbers to generate. In this scenario, `n` is the number of ignitions to simulated per Monte Carlo run.
- `meanlog` is the mean of the logarithm of the distribution with default set to 0.
- `sdlog` is the standard deviation of the logarithm of the distribution with default set to 1.

Note: You should always use the log-scale parameters of mean and SD in `rlnorm()`.

1.4 Task 1

□ Goal: Start by setting a seed. Generate a large vector of simulated 8-hour distances for the **High** category using `rlnorm()`. Look up the **High** row's **Mean log** and **SD log** in **Table 1**, and store the result as `High_Lognorm`. Set the number of ignitions per season to 450.

```
set.seed(12345)
# □ Test
High_Lognorm <- rlnorm(n = 450, meanlog = 1.907, sdlog = 0.280)
```

1.5 Task 2

□ / □ Checks and Comment

Summarize what you just simulated. Report the median, mean, and 90th percentile of `High_Lognorm`, and comment in one sentence on the right-skew you expect for a lognormal.

```
# □ Verify
median(High_Lognorm)
```

```
## [1] 7.035328
```

```
mean(High_Lognorm)
```

```
## [1] 7.224904
```

```
quantile(High_Lognorm, 0.9)
```

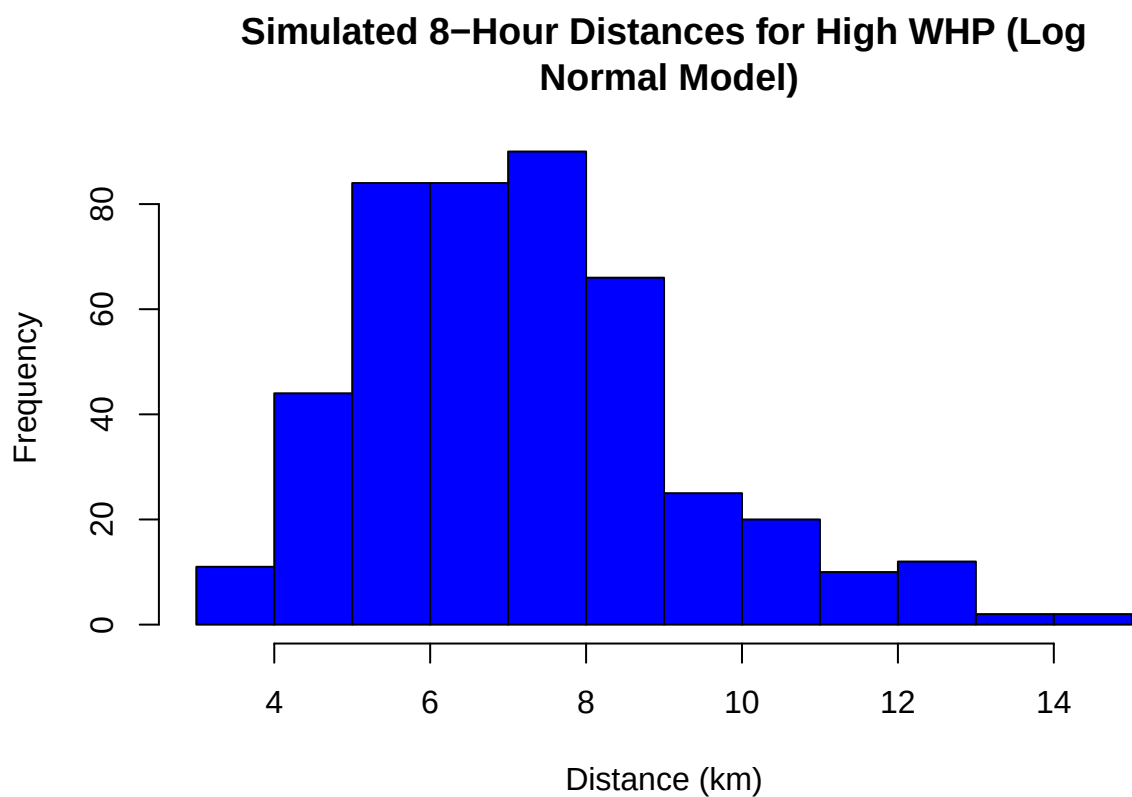
```
## 90%
## 10.03975
```

1.6 Task 3

□ / □ Checks and Comment

Make a histogram of High_Lognorm. In one sentence, note how the shape matches your summary in Task 2.

```
# □ Verify
hist(High_Lognorm,
     col = "blue",
     main = "Simulated 8-Hour Distances for High WHP (Log
Normal Model)",
     xlab = "Distance (km)")
```



1.7 Task 4

□ Goal: Copy the code that created High_Lognorm and nest it inside of the mean() function. Set the rlnorm() call to >= 10. This one line will compute the proportion of a single sample that reaches or exceeds a threshold of 10 km spread in 8-hours. Store this result as p_high_one

```
# □ Test
p_high_one <- mean(rlnorm(n = 450, meanlog = 1.907, sdlog = 0.280) >= 10) # one season, 450 ignitions
```

1.8 Task 5

□ / □ Checks and Comment

Comment about meaning of `p_high_one`. What does this value represent in the context of wildfire spread risk for ignitions in the **High** WHP category?

```
# □ Verify  
p_high_one
```

```
## [1] 0.07333333
```

1.9 Task 6

□ Goal: Turn that single-season calculation into many seasons. Use `replicate()` to repeat the same inside expression for 2,000 seasons. Store the resulting vector of proportions as `p_high_many`, then report its mean and make a small histogram of those season-level proportions.

```
# □ Test  
p_high_many <- replicate(  
  n = 2000, # 2,000 seasons  
  expr = mean(rlnorm(n = 450, meanlog = 1.907, sdlog = 0.280) >= 10) # 450 wildfires, 10 km threshold  
)
```

```
# □ Verify  
mean(p_high_many)
```

```
## [1] 0.07926556
```

```
mean(p_high_many >= 1)
```

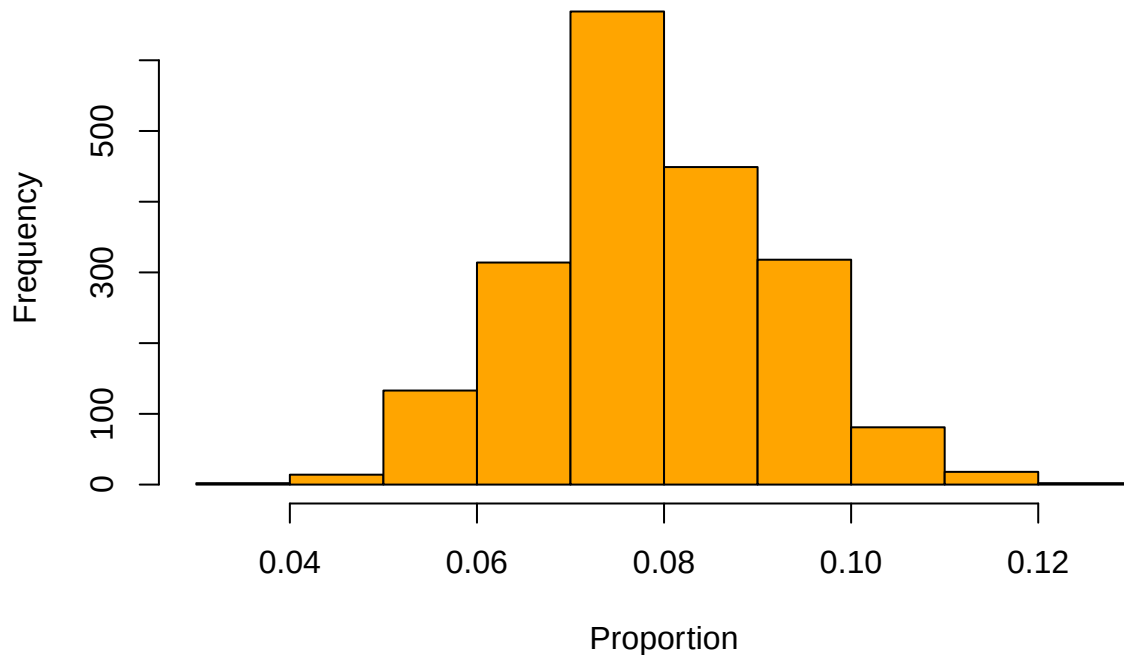
```
## [1] 0
```

1.10 Task 7

□ / □ Checks and Comment

```
# □ Verify  
hist(p_high_many,  
  col = "orange",  
  main = "Season-level proportions  $\geq k$  (High)",  
  xlab = "Proportion")
```


Season-level proportions = k (High)



1.11 Task 8

□ Goal: Create four parallel vectors from Table 1: `cats`, `prob` (as proportions that sum to 1), `meanlog`, and `sdlog` in matching order. Build the inside expression for a single season of $N <- 450$ ignitions: first sample a category index for each ignition using `sample()` with `prob`, then draw distances with `rlnorm()` using the matched `meanlog` and `sdlog`. Return a count of ignitions $\geq k$. Run it once to confirm it works.

```
# □ Test
cats <- c("Very Low", "Low", "Moderate", "High", "Very High", "Extreme", "Ultra")
prob <- c(0.25, 0.25, 0.25, 0.15, 0.05, 0.04, 0.01) # proportions sum to 1
meanlog <- c(0.582, 1.046, 1.451, 1.907, 2.455, 2.866, 3.310)
sdlog <- c(0.472, 0.325, 0.325, 0.280, 0.246, 0.220, 0.212)
N <- 450 # ignitions per season
one_mixture_count <- {
  idx <- sample.int(length(cats), size = N, replace = TRUE, prob = prob)
  d <- rlnorm(N, meanlog = meanlog[idx], sdlog = sdlog[idx])
  sum(d >= 10)
}
one_mixture_count
```

```
## [1] 34
```

1.12 Task 9

□ Goal: Wrap the mixture inside code from Task 7 in `replicate()` for S seasons. Save the season counts as `counts_mix`. Report the expected number of $\geq k$ runs and the probability that a season has more than 10 such runs (change the threshold if instructed).

```
# □ Test
counts_mix <- replicate(
  n = 2000,                                # 2,000 seasons
  expr = {
    idx <- sample.int(length(cats), size = N, replace = TRUE, prob = prob)
    d <- rlnorm(N, meanlog = meanlog[idx], sdlog = sdlog[idx])
    sum(d >= 10)
  }
)
```

1.13 Task 10

□ / □ Checks and Comment

```
# □ Verify
mean(counts_mix)      # expected number of  $\geq k$  runs
```

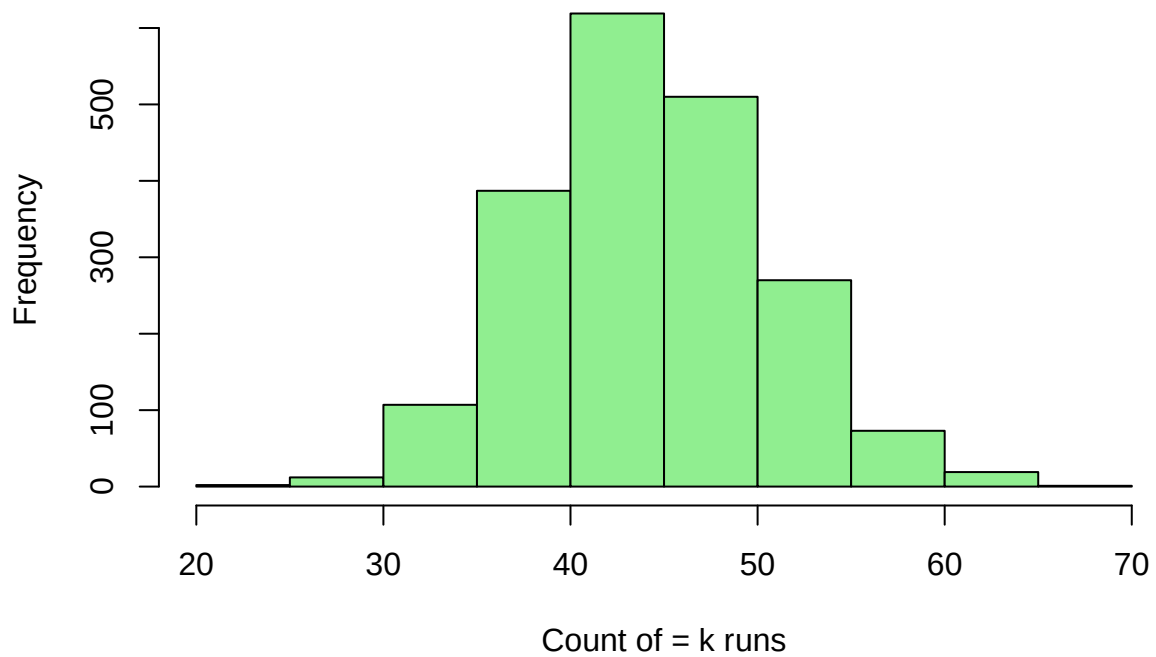
```
## [1] 44.7605
```

```
mean(counts_mix > 10) # probability of  $> 10$  such runs
```

```
## [1] 1
```

```
# □ Verify
hist(counts_mix,
  col = "lightgreen",
  main = "Season-level counts of  $\geq k$  runs (Mixture Model)",
  xlab = "Count of  $\geq k$  runs")
```

Season-level counts of = k runs (Mixture Model)



1.14 Reflections

□ Comment: In 1-2 sentences, reflect on what you learned from this assignment about using simulation to estimate wildfire spread risk. Mention any challenges you faced and how you overcame them.

2. Save and Upload

1. You will be submitting **both** the *Your Pick* Document and the workspace file. The workspace file saves all the objects in your environment that you created in this lesson. You can save the workspace by running the following command in a code chunk of the Quarto Document document:

```
save.image("Assignment12_Workspace.RData")
```

Or you can click the “Save Workspace” button in the Environment pane.

□ **Always save the R documents before closing.**

2. Find the assignment in this week’s module in Canvas and upload **both** the RMD and the workspace file.

3. Today you practiced:

- Used `sample()` for discrete simulations with or without replacement.
- Visualized uniform and aggregated dice results with histograms.
- Generated normal random numbers with `rnorm()` and plotted the shape.
- Repeated an experiment easily with `replicate()`.
- Compared two simulated distributions to spot bias.